21/SC/MA/285

UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
SECOND SEMESTER DEGREE EXAMINATION - 2022/2023 SESSION

MTH 224: LINEAR ALGEBRA II

INSTRUCTION: ANSWER ANY FOUR QUESTIONS

TIME: 2.5 HOURS

1(a) When are two matrices A and B said to be similar?

(b) Find the eigenvalues and eigenvectors of the transformation $A = \begin{pmatrix} -2 & -4 & 2 \\ -2 & 1 & 2 \\ 4 & 2 & 5 \end{pmatrix}$.

characteristics Polynomial

2(a) Diagonalize the matrix $B = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$.

$P(\lambda) = \lambda^3 - Tr(A)\lambda^2 + (m_{11}+m_{22}+m_{33})\lambda - |A| = 0$

3 (a) Consider V as vector space. Let $S_1 = \{(1,1,-1),(1,-1,0),(0,1,2)\}$ and $S_2 = \{(1,2,1),(2,0,-1),(-2,3,2)\}$ be two different bases of V. Obtain the transition matrix from S_1 to S_2.

(b) Solve the system of linear equations using the adjoint method:

$$x_1 - 2x_2 + 3x_3 = 9$$
$$-x_1 + 3x_2 - 5x_3 = -6$$
$$2x_1 - 5x_2 + 5x_3 = 17$$

$4 \times 4 = 72$

4(a) State Caley Hamilton's theorem.

(b) Show that the transformation $A = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}$ fulfills Caley Hamilton's theorem.

if $\lambda^3 + \lambda^2 - 12\lambda = 0$

5(a) When is a matrix P said to be orthogonal?

(b) Is the matrix $P = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ 2 & 1 & -2 \\ 2 & 2 & -1 \end{pmatrix}$ is orthogonal?

if $A \cdot A^T = I$

(c) Establish that similar matrices have the same characteristic polynomial.

λ

$-27-12-150$

$125-100-125$

$125-100-$

$\lambda = 5$

λ

16×3